

A Quick Look at File System Partitioning and Logical Volume Management



Disk partitioning facilitates better utilisation and management of disk space. This article goes into the basics of disk management. It also is a guide to logical volume management.

Disk partitioning allows a hard drive to be divided into multiple logical storage units that are referred to as partitions. By dividing the disk into partitions, systems administrators can use different partitions to perform different functions.

The MBR partitioning scheme

This scheme supports a maximum of four primary partitions. On Linux systems, with the use of extended and logical partitions, administrators can create a maximum of 15 partitions. Since the data in the partitions is stored as 32-bit values, disks partitioned with the MBR (Master Boot Record) scheme have a maximum disk and partition size limit of 2TiB.

Creating MBR disk partitions

For disks with the MBR partitioning scheme, the `fdisk` partition editor can be used to create disk partitions. Creating an MBR style disk partition involves the following steps.

Specify the disk device to create the partition on

As the root user, execute the `fdisk` command and specify the disk device's name as an argument. This will start the `fdisk` command in interactive mode, and will present a command prompt:

```
# fdisk /dev/vdb
```

Request a new primary or extended partition

Enter `n` to request a new partition, and specify whether the partition should be created as a primary or extended partition. The default selection is the primary type.

Note: For situations in which more than four partitions are needed on a disk, this limit can be bypassed by creating three primary partitions and one extended partition. This extended partition serves as a container within which multiple logical partitions can be created.

Specify the partition number

This serves as the identification number of the new partition on the disk for use in future partition operations. The default value is the lowest unused partition number.

Specify the first sector on the disk that the new partition will start on

The default value is the first sector available on the disk.

Specify the last sector on the disk that the new partition will end on

The default value is the last of the available, unallocated sectors contiguous to the new partition's first sector.

In addition to the ending sector number, `fdisk` also accepts a number representing the desired size of the partition expressed in